431 Lab 3 Instructions

Fall 2026 - deadline in Course Calendar

Thomas E. Love

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! Important

- This Lab contains 4 tasks for you to complete.
- The deadline for completing this Lab is posted in the Course Calendar.

0.1 R Setup

```
knitr::opts_chunk$set(comment = NA)

library(tidyverse)

theme_set(theme_bw())
```

You will use additional packages in developing your response[1].

We have built no template for Lab 3. Instead, use the templates for Lab 1 and Lab 2 we have provided to help you get started.

0.2 Learning Objectives for this Lab

1. Reflect meaningfully on how the PPDAC cycle might be used in scientific or other practical work.
2. Ingest data from an Excel (.xlsx) file into R
3. Compare distributions of three independent samples visually and numerically.
4. Build an appropriate analysis of variance model, and use it to motivate comparisons overall and across pairs of groups.
5. Communicate effectively about your results in complete, clear English sentences.

0.3 Getting Help

You may discuss each Lab with Professor Love, the teaching assistants or your colleagues, but your answer must be prepared by **you working alone**. Don't be afraid to ask questions, using any of the methods described on our Contact Us page.

[1]In my answer sketch, I used janitor, knitr, DescTools, readxl and easystats in addition to the tidyverse. I also loaded the Love-431.R script.

1 Task 1 (10 points)

Write a short essay (of 150-300 words, please) describing how the PPDAC problem solving cycle (first described by Spiegelhalter in his introduction) might be helpful to you in the context of some "problem" you are interested in solving.

Please feel free to draw on your own experience solving problems in a systematic way, and don't feel obliged to write about a "problem" that is related to biology or medicine or health or science. Anything you can explain briefly and that you are interested in could work well here.

Your response should be written using clear and complete English sentences and minimizing jargon.

2 Task 2 (20 points)

2.1 The Data

Tasks 2 and 3 use data gathered in the `lab3.xlsx` Excel file, which we have simulated to reflect a hypothetical clinical trial. In this trial, the investigators are testing out a new drug to see its effect on a subject's systolic blood pressure (SBP). For our purposes, if a subject's SBP is over 130 mm Hg they are considered to have hypertension. The goal was to reduce the SBP from a baseline level by using a new drug (Treatment C), and comparing that to the current top-of-the-line drug (Treatment B), and the oldest drug (Treatment A). The trial focused specifically on non-Hispanic African-American women who were in long-term relationships and between the ages of 55 and 65 years old, with a minimal comorbidity profile.

The outcome of interest is the post-treatment systolic blood pressure (`sbp_follow`), and we are also given the subject's age, their pre-treatment systolic blood pressure (`sbp_baseline`) and whether or not the subject's partner has hypertension.

Variable	Description
`subjectid`	unique subject identifier
`group_rx`	treatment group where 1 = Group A, 2 = Group B, 3 = Group C
`partner`	whether or not the subject's partner also has hypertension
`age`	subject age in years at baseline
`sbp_baseline`	subject's baseline systolic blood pressure (mm Hg)
`sbp_follow`	subject's follow-up systolic blood pressure (mm Hg)

2.2 The Task

Ingest the data, and make `group_rx` into a factor variable called `group_f` which has levels "Group_A", "Group_B", and "Group_C" as appropriate.

Then produce at least two visualizations comparing the three treatment groups. Your goal should be to assess the assumptions of an analysis of variance which compares the SBP at follow-up (`sbp_follow`) means across the `groups`, ignoring all other information in the data.

Then, produce a numerical summary for `sbp_follow` within each group that includes, at minimum, the following summaries:

- sample size, mean, standard deviation, median, MAD, minimum and maximum

After you have completed this coding, write a paragraph describing your results. Specifically, your paragraph should tell us what conclusions can you draw about the center, spread and shape of the data in each group, and what you conclude about ANOVA model assumptions in light of your summaries.

2.2.a Two Hints for Task 2

1. An appropriate way to describe center, spread and shape involves a graph (or set of them) that show us the distribution of the outcome within each group. There are lots of available options. You could compare (a) histograms, or compare (b) boxplots, or compare (c) Normal Q-Q plots, for example. You might want to look at more than one of these options, but I wouldn't think you would need all three, though. Don't include plots that you're not going to discuss in your paragraph describing your results.

2. The primary ANOVA assumptions that we're looking at in Task 2 are linearity, homogeneity of variance, and Normality. You could do this with `check_model()` on the linear model supporting your ANOVA, or you could do this without `check_model()` through the combination of (a) a discussion at the plot of the `sbp_follow` data by `group` you developed earlier in Task 2, and (b) a Normal Q-Q plot of the residuals from the linear model supporting your ANOVA. Again, though, you shouldn't include plots that you're not going to discuss in your description of the results.

3 Task 3 (15 points)

Now complete the comparison of the SBP at follow-up means of the three treatment groups (A, B and C) described in Task 2 using an analysis of variance, backed by an OLS linear model.

What conclusions do you draw about the groups (a) overall, and (b) in appropriate pairwise pre-planned comparisons[2], using a **90%** confidence level?

Write a paragraph to describe your results, using clear and complete English sentences and minimizing jargon.

4 Task 4 (5 points)

Share something useful that you learned from reading the introduction and sections 1-11 of R for Data Science, including a specific quote from the book.

[2] A Tukey HSD approach is appropriate here.

Your response should include a useful quote or other reference from the book (specify the section number for the quote), plus two or three clear and complete English sentences that tell us why you found this to be particularly helpful.

5 Section 5 of your Lab Report: AI Usage

All students should include an AI Usage section in each assignment for this class. See the instructions from Lab 1.

6 Section 6 of your Lab Report: Session Information

At the end of your Quarto file, you should run session information. The code required is shown below.

```
xfun::session_info()
```

7 Additional Notes and Instructions

7.1 Submitting this Lab

Submit this Lab via Canvas, using the Lab 3 assignment. Be sure to submit both files:

1. Your Quarto file (.qmd).
2. The HTML file you obtain by knitting the Quarto file (.html)

7.2 Grading this Lab

This Lab will be graded by the TAs and then reviewed by Dr. Love. Your grades will be available one week after the Lab deadline.

The maximum score on this Lab is 50 points.

As each Lab passes its deadline (as listed in the Course Calendar), we will:

- post the answer sketch (48 hours after the deadline) and draft grading rubric to our Shared Google Drive, and then
- post grades and any revisions to the grading rubric or answer sketch one week after the deadline to a location we will provide to you.

7.3 Emergencies and Late Policy

We do not grant extensions on Lab deadlines.

- To receive full credit on a Lab, it must be received on Canvas no later than 59 minutes after the posted deadline. (This allows for small issues with uploading to Canvas to occur without penalty.)
  - Labs that are turned in 1-48 hours after the deadline will lose 10 points for late work.
- No extensions to Lab deadlines will be made this semester. Labs turned in more than 48 hours after the deadline will receive no credit, since by then the Lab Sketch will be posted.
- For details on how Lab grades affect your course grade, see the Syllabus.
- If you have an emergency that will keep you from submitting the Lab by even the late deadline of Friday at noon, please let Dr. Love know that (as soon as possible) via email and he will consider excusing you from the Lab.

7.4 Lab Regrade Requests

If, after your Lab is graded, you want Dr. Love to review the grading or correct a grading error, please follow the Lab Regrade Request policy posted on our Labs page.

8 Session Information

```r
xfun::session_info()
```

```
R version 4.6.1 (2026-06-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)
```

```
Locale:
  LC_COLLATE=English_United States.utf8
  LC_CTYPE=English_United States.utf8
  LC_MONETARY=English_United States.utf8
  LC_NUMERIC=C
  LC_TIME=English_United States.utf8

Package version:
  askpass_1.2.1        backports_1.5.1     base64enc_0.1.6
  bit_4.6.0            bit64_4.8.2         blob_1.3.0
  broom_1.0.13         bslib_0.11.0        cachem_1.1.0
  callr_3.8.0          cellranger_1.1.0    cli_3.6.6
  clipr_0.8.1          compiler_4.6.1      conflicted_1.2.0
  cpp11_0.5.5          crayon_1.5.3        curl_7.1.0
  data.table_1.18.4    DBI_1.3.0           dbplyr_2.6.0
  digest_0.6.39        dplyr_1.2.1         dtplyr_1.3.3
  evaluate_1.0.5       farver_2.1.2        fastmap_1.2.0
  fontawesome_0.5.3    forcats_1.0.1       fs_2.1.0
  gargle_1.6.1         generics_0.1.4      ggplot2_4.0.3
  glue_1.8.1           googledrive_2.1.2   googlesheets4_1.1.2
  graphics_4.6.1       grDevices_4.6.1     grid_4.6.1
  gtable_0.3.6         haven_2.5.5         highr_0.12
  hms_1.1.4            htmltools_0.5.9     httr_1.4.8
  ids_1.0.1            isoband_0.3.0       jquerylib_0.1.4
  jsonlite_2.0.0       knitr_1.51          labeling_0.4.3
  lifecycle_1.0.5      lubridate_1.9.5     magrittr_2.0.5
  memoise_2.0.1        methods_4.6.1       mime_0.13
  modelr_0.1.11        openssl_2.4.2       otel_0.2.0
  pillar_1.11.1        pkgconfig_2.0.3     prettyunits_1.2.0
  processx_3.9.0       progress_1.2.3      ps_1.9.3
  purrr_1.2.2          R6_2.6.1            ragg_1.5.2
  rappdirs_0.3.4       RColorBrewer_1.1-3  readr_2.2.0
  readxl_1.5.0         rematch_2.0.0       rematch2_2.1.2
  reprex_2.1.1         rlang_1.3.0         rmarkdown_2.31
  rstudioapi_0.19.0    rvest_1.0.5         S7_0.2.2
  sass_0.4.10          scales_1.4.0        selectr_0.6.0
  stats_4.6.1          stringi_1.8.7       stringr_1.6.0
  sys_3.4.3            systemfonts_1.3.2   textshaping_1.0.5
  tibble_3.3.1         tidyr_1.3.2         tidyselect_1.2.1
  tidyverse_2.0.0      timechange_0.4.0    tinytex_0.60
  tools_4.6.1          tzdb_0.5.0          utf8_1.2.6
  utils_4.6.1          uuid_1.2.2          vctrs_0.7.3
```

```
viridisLite_0.4.3   vroom_1.7.1         withr_3.0.3
xfun_0.60           xml2_1.6.0          yaml_2.3.12
```